

AI Travel Planner Business Use Case

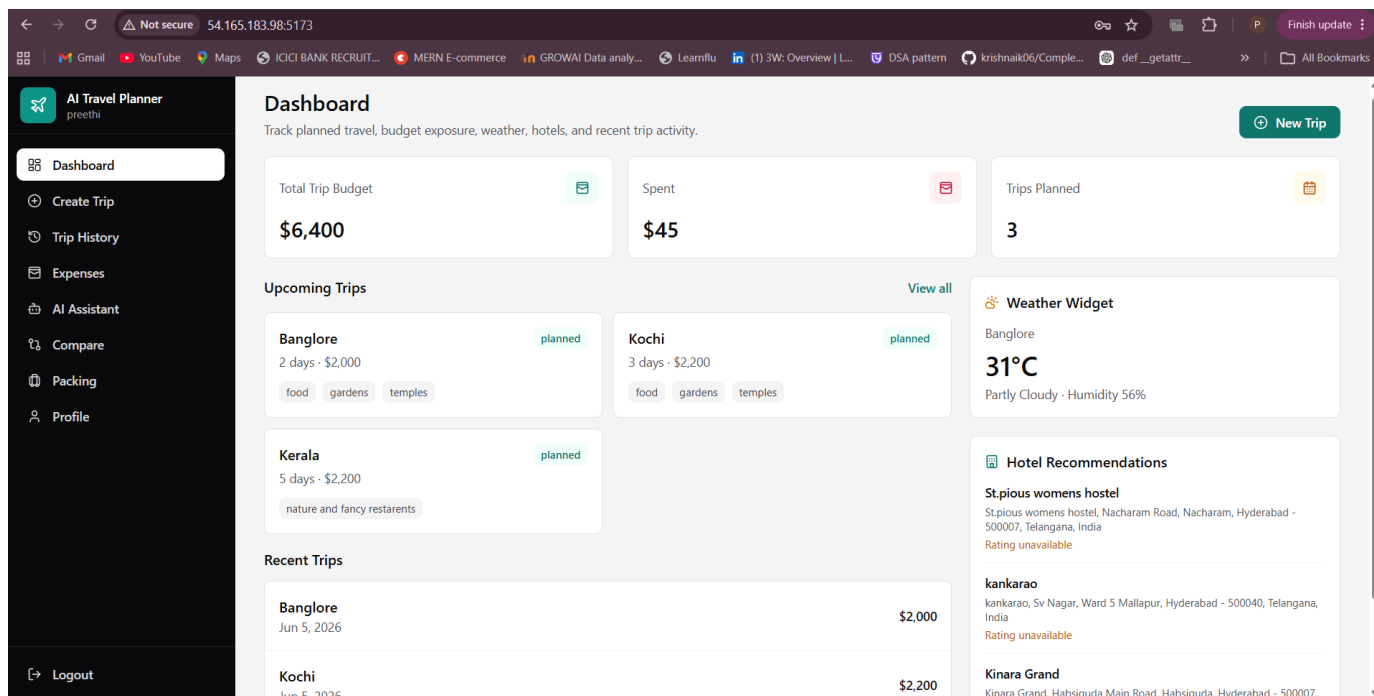
Business Requirement

AI Travel Planner is a cloud-native travel management platform that helps users plan trips, generate AI-powered itineraries, manage travel expenses, compare destinations, and receive travel recommendations.

The application leverages Artificial Intelligence to create personalized travel plans based on destination, budget, trip duration, and user interests. It also assists travelers with destination comparison, expense tracking, packing recommendations, and travel guidance.

Primary Business Value:

- Generate AI-powered travel itineraries
- Track and manage travel expenses
- Compare destinations using weather, activities, and cost data
- Provide intelligent travel assistance through an AI chatbot
- Generate destination-specific packing checklists
- Store travel history and planned trips
- Secure application secrets using AWS Secrets Manager
- Encrypt sensitive application data using AWS KMS
- Archive application logs into Amazon S3 for auditing and troubleshooting
- Deploy a scalable cloud-native application on AWS EC2



Application Features

1. AI Trip Generation

Users can generate personalized travel itineraries based on destination, budget, duration, and interests. The AI engine produces day-wise travel plans, budget breakdowns, and travel recommendations

The screenshot shows the 'Create Trip' page of the AI Travel Planner. The left sidebar contains navigation links: Dashboard, Create Trip (active), Trip History, Expenses, AI Assistant, Compare, Packing, and Profile. The main content area is titled 'Create Trip' with the subtitle 'Generate an AI itinerary and save the trip to your travel history.' It features input fields for Destination (kochi), Budget (5000), Days (2), and Interests (temples, food, trekking). A green 'Generate AI Trip' button is at the bottom. The right panel displays the generated itinerary summary, including destination, duration, budget, interests, a day-wise plan for two days, a budget breakdown table, and travel tips.

Category	Amount
Accommodation	1500
Food	1000
Transportation	500
Activities	1000

Tip
Best Time To Visit: September to February
What To Wear: Light and comfortable clothing
Local Transportation: Use autos or buses for local travel

2. Trip History Management

Users can review previously planned trips, update trip status, and manage travel records.

The screenshot shows the 'Trip History' page of the AI Travel Planner. The left sidebar is the same as the previous page. The main content area is titled 'Trip History' with the subtitle 'Review trips, update status, and remove trips that no longer belong in your plan.' It displays a list of four trips: Kochi, Bangalore, Kochi, and Kerala. Each trip entry shows the destination, duration, budget, creation date, status (planned), and a delete button.

Destination	Duration	Budget	Created	Status	Action
kochi	2 days	\$5,000	Created Jun 5, 2026	planned	Delete
Banglore	2 days	\$2,000	Created Jun 5, 2026	planned	Delete
Kochi	3 days	\$2,200	Created Jun 5, 2026	planned	Delete
Kerala	5 days	\$2,200	Created Jun 2, 2026	planned	Delete

3. Expense Tracking

The application allows travelers to record expenses and compare actual spending against planned budgets.

The screenshot shows the 'Expense Tracker' page of the 'AI Travel Planner' application. The left sidebar contains navigation links: Dashboard, Create Trip, Trip History, Expenses (selected), AI Assistant, Compare, Packing, and Profile. The main content area has a header 'Expense Tracker' with a sub-header 'Log trip spending by category and compare it against your planned budget.' Below this is a form to add a new expense with fields for Trip (Bangalore), Category (Transport), Amount, and Description. A 'Save Expense' button is at the bottom. To the right, a section titled 'Bangalore Expenses' shows a trip budget of \$2,000 and a total spent of \$545. It lists two expenses: Transport (\$500) and Food (\$45).

Category	Amount
Transport	\$500
Food	\$45

4. AI Travel Assistant

Integrated AI assistant helps users with travel planning, budgeting, routing, weather information, and destination recommendations.

The screenshot shows the 'AI Travel Assistant' page of the 'AI Travel Planner' application. The left sidebar is the same as in the previous screenshot, with 'AI Assistant' selected. The main content area has a header 'AI Travel Assistant' with a sub-header 'Ask trip planning, routing, safety, budget, and timing questions.' Below this is a chat interface with a message from the assistant: 'Tell me where you are going, your dates, budget, and what you want to optimize.' At the bottom, there is a text input field with a placeholder 'Ask about routes, budgets, hotels, packing, or weather' and a send button.

5. Destination Comparison

Users can compare destinations using weather conditions, activity availability, and travel costs to identify the most suitable destination.

The screenshot shows the 'Destination Comparison' page of the AI Travel Planner. The left sidebar contains navigation links: Dashboard, Create Trip, Trip History, Expenses, AI Assistant, Compare (selected), Packing, and Profile. The main content area is titled 'Destination Comparison' with a subtitle 'Compare destinations across costs, weather, activities, and final fit.' It features two input fields for 'Destination A' (banglore) and 'Destination B' (kochi), followed by a 'Compare' button. Below this, a 'Best Choice' section highlights 'banglore'. The interface is divided into three columns: 'Cost' (accommodation: ['banglore': 30, 'kochi': 25], food: ['banglore': 15, 'kochi': 10], transportation: ['banglore': 20, 'kochi': 15]), 'Weather' (temperature: ['banglore': 28, 'kochi': 32], humidity: ['banglore': 60, 'kochi': 80]), and 'Activities' (outdoor_activities: ['banglore': 8, 'kochi': 10], cultural_events: ['banglore': 12, 'kochi': 8]).

6. Packing Assistant

Generates customized packing checklists based on destination and travel month.

The screenshot shows the 'Packing Assistant' page of the AI Travel Planner. The left sidebar is identical to the previous page, with 'Packing' selected. The main content area is titled 'Packing Assistant' with a subtitle 'Build a month-aware packing checklist for your destination.' It features two input fields for 'Destination' (kochi) and 'Month' (may), followed by a 'Build List' button. Below this, the interface is divided into five columns: 'clothing' (t-shirts, lightweight pants, shorts, dresses, umbrella), 'toiletries' (toothbrush, toothpaste, shampoo, conditioner, sunscreen), 'electronics' (phone, laptop, power bank, camera, universal power adapter), 'travel documents' (passport, visa, flight tickets), and 'miscellaneous' (water bottle, snacks, first aid kit). Each item has a checkbox for selection.

Deployment Architecture

The AI Travel Planner application is deployed on an Amazon EC2 instance.

Application Components:

- Frontend – React
- Backend – FastAPI
- Database – PostgreSQL
- Secrets Management – AWS Secrets Manager
- Encryption – AWS KMS
- Log Storage – Amazon S3
- Hosting Platform – Amazon EC2

The application was successfully deployed and tested end-to-end, including trip generation, expense tracking, destination comparison, AI assistant functionality, secret retrieval, KMS integration, and S3 log archival.

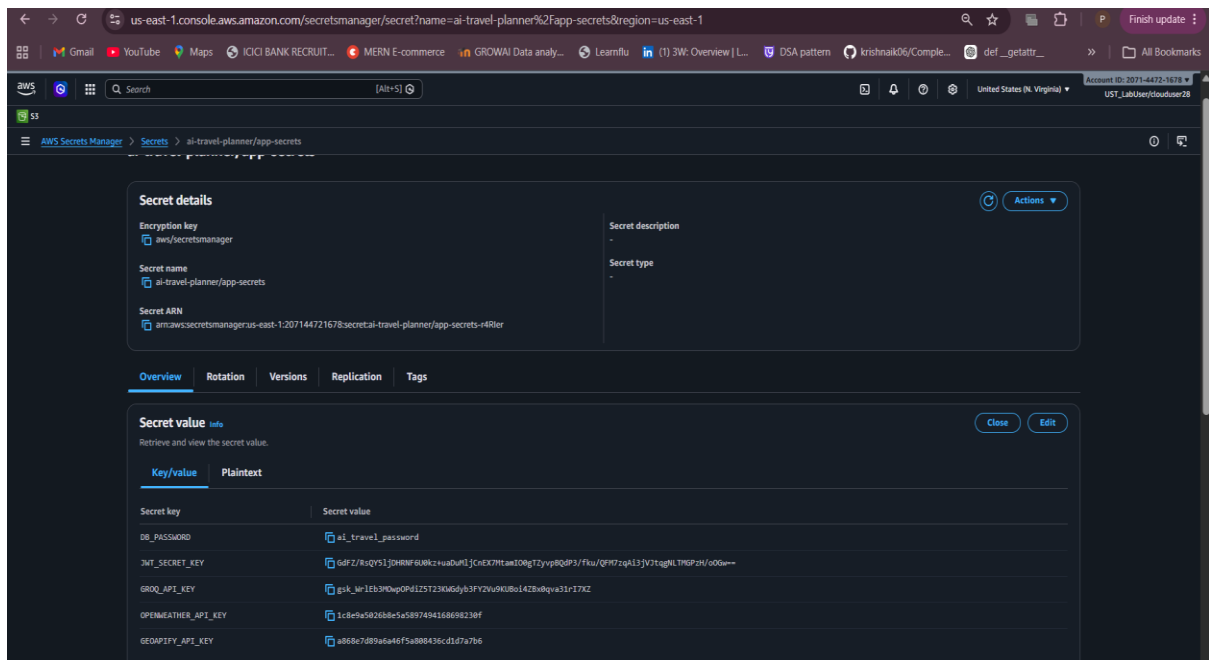
AWS Services Used

PostgreSQL

PostgreSQL is used as the primary database for storing user accounts, trip history, expense records, destination comparisons, and itinerary information.

The database provides structured storage and allows users to retrieve travel plans even after application restarts.

AWS Secrets Manager



```
ubuntu@ip-172-31-81-13:~/New-AI/AI-Travel_Planner$ cat .env
ENVIRONMENT=local

AWS_REGION=us-east-1

SECRET_NAME=ai-travel-planner/app-secrets

JWT_ALGORITHM=HS256

ACCESS_TOKEN_EXPIRE_MINUTES=1440

CORS_ORIGINS=http://54.165.183.98:5173,http://54.165.183.98:8080

VITE_API_BASE_URL=http://54.165.183.98:8080/api
GEOPIFY_API_KEY=a868e7d89a6a46f5a808436cd1d7a7b6
GOOGLE_MAPS_API_KEY=
ubuntu@ip-172-31-81-13:~/New-AI/AI-Travel_Planner$
```

i-054d80c694096cbde (AI-traveller)

PublicIPs: 54.165.183.98 PrivateIPs: 172.31.81.13

Implemented AWS Secrets Manager to securely store and retrieve application secrets and configuration values.

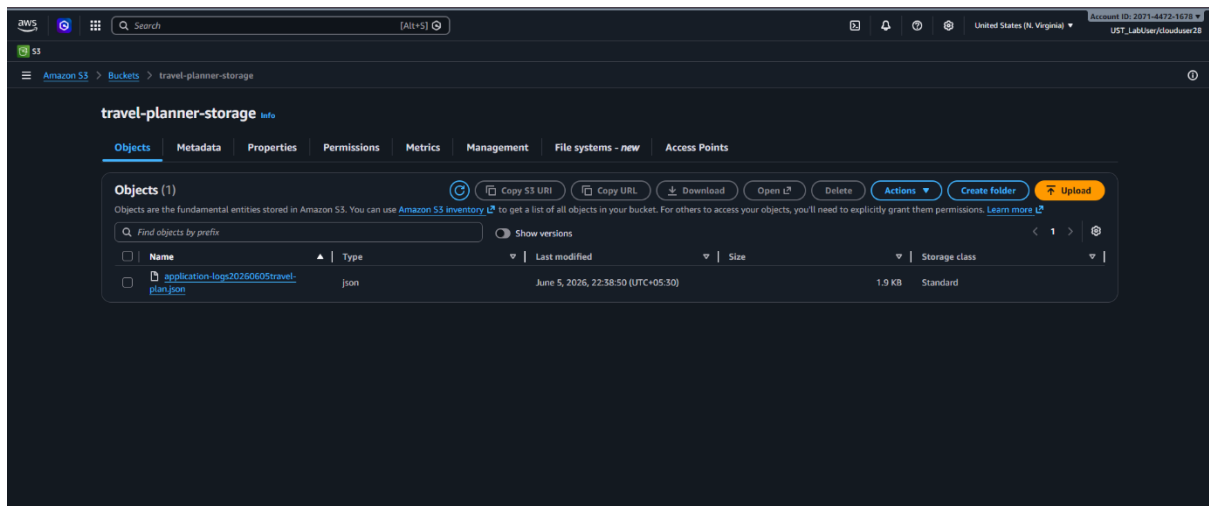
The following sensitive information is stored securely in AWS Secrets Manager:

- Database Password
- JWT Secret Key
- GROQ API Key
- OpenWeather API Key
- Geoapify API Key

The application retrieves secrets dynamically during runtime using AWS SDK integration. No sensitive credentials are hardcoded in the source code.

This approach improves application security and simplifies secret management across environments.

Amazon S3 Log Storage



```
travel-planner-storage
application-logs20260605travel-plan.json
```

Implemented Amazon S3 as a centralized storage location for application logs.

The application periodically archives operational logs and activity records into an S3 bucket named travel-planner-storage.

Benefits of S3 log archival include:

- Long-term log retention
- Centralized log management
- Audit and compliance support
- Cost-effective storage
- Disaster recovery support

Log files are stored as JSON objects within the S3 bucket, enabling future analysis and troubleshooting of application events.